Do Attributions Tell the Truth About Data Filters?

Measuring the Faithfulness of SHAP and LIME in Identity-Term Bias Audits

Julia Hooper, Bláithín Kavanagh, Rory Linnane and April Gilhool
University of Limerick

***Abstract*—Pretraining data filters silently decide whose language large language models learn from, and have been shown to remove text from specific groups. They are audited only in aggregate, and the feature-attribution methods an auditor would use to explain individual removals have not been tested for faithfulness. We evaluate SHAP and LIME on the Detoxify filter with four tests: (i) agreement with counterfactual identity-word swaps, (ii) agreement with exact Shapley values on a transparent filter, (iii) a deletion test, and (iv) a stability test. Both explainers rank identity-word effects almost as the counterfactual does (Spearman $\rho \geq 0.92$ for SHAP, 0.97 for LIME) and recover exact Shapley values ($\rho \geq 0.97$), but SHAP overstates the size of the effect, and its error depends on how removed words are replaced. On weak signals LIME's surrogate fit falls (R^2 0.77 to 0.41), and SHAP–LIME disagreement is no better than chance (AUC 0.46) at flagging wrong explanations. Attribution can therefore triage a filter for identity-term bias, but cannot certify it as fair.**

***Index Terms*—explainable AI, feature attribution, SHAP, LIME, faithfulness, pretraining data filters, social bias**

I. Introduction

Data filters decide whose text large language models learn from. C4's blocklist, for example, removed 42% of documents classified as African American English but only 6.2% of those classified as White-aligned English [1]. The same study found that documents mentioning sexual orientation were among the most likely to be dropped. Quality and toxicity classifiers show related patterns [2], [3]. However, such filters are audited only in aggregate: we learn which groups lose text, but not why any single document was removed. A blocklist's reason for removal is the listed word it matched, whereas a learned filter such as Detoxify [4] or HarmFormer [5] gives no reason.

An auditor would therefore turn to post-hoc feature attribution, chiefly SHAP [6] and LIME [7]. However, a post-hoc explanation only approximates the model and can be wrong [8], and explainers often disagree with one another [9]. Agreement between two explainers does not make either correct. Before an audit relies on them, they must be shown to be *faithful*, meaning they reflect what the filter did [10].

We ask: **are SHAP and LIME faithful enough to support identity-term bias audits of a pretraining data filter, and when do they stop being trustworthy?** We test them on Detoxify against references whose correct answers we can compute independently. Our contributions are: (i) a counterfactual reference built from minimal-pair identity-word swaps; (ii) an exact-Shapley reference from a transparent filter; (iii) a deletion-based comparison of SHAP, LIME and random words; and (iv) a test of whether simple cues can warn an auditor that an explanation is unreliable. The cues are signal strength, LIME's fit quality and SHAP–LIME disagreement. We find that attribution is faithful when the identity signal is strong and degrades when it is weak, so it can flag identity-term bias but cannot certify a filter as fair.

II. Background and Related Work

Bias in Pretraining Data Filters. Beyond blocklist and classifier biases (Sec. I), filters also misread context. Mendu et al. found that a transformer-based filter flags news articles that *report* on harmful content [5]. Udagawa et al. found that a fairness-oriented regard classifier removes sentences reporting real-world events involving marginalised groups [11]. Dixon et al. isolated unintended classifier bias with minimal-pair templates that swap one identity word for a neutral control (e.g. "my neighbour is *gay*" vs. "*tall*") [12], a design we reuse.

Feature Attribution for Text. SHAP attributes a prediction using Shapley values, the unique attribution satisfying efficiency, symmetry and the null-player axiom [6]. For text, the SHAP library uses a partition approximation; exact values can be computed only for short sentences and serve as our reference (Sec. III). LIME instead fits a locally weighted sparse linear surrogate to perturbations of the input, with the surrogate's R^2 measuring local fidelity [7]. Both are post-hoc, local and model-agnostic, needing only query access [13], [14].

Evaluating Explanations: Faithfulness vs. Agreement. An explanation is *faithful* if it reflects the model's actual reasoning, which is distinct from looking plausible to a human [10]; our tests are *functionally grounded*, using formal proxies rather than human studies [15]. ERASER's deletion metrics, comprehensiveness and sufficiency [16], are one such proxy, but deleting words creates out-of-distribution inputs, so score changes may partly reflect distribution shift [17]. Explainers also disagree with one another [9], can be manipulated to hide discriminatory behaviour [18], and are not necessarily most faithful on the most accurate classifier [19]. These evaluations used general classification tasks; to our knowledge, no prior work has evaluated attribution faithfulness in pretraining filter audits against references whose correct answers are known independently.

III. Method

Filters. We audit two Detoxify toxicity classifiers [4]: `original`, a BERT model trained on the Jigsaw Toxic Comment Classification Challenge, and `unbiased`, a RoBERTa model trained on Civil Comments for the Jigsaw Unintended

Bias challenge [20]. In Exp 4 we train a transparent filter: a logistic regression on binary word and word-pair features, fitted to 100,000 Civil Comments, each labelled toxic if at least half of its annotators rated it toxic. On 5,000 held-out comments its AUC is 0.892, against 0.928 for `original` and 0.971 for `unbiased`, which has an advantage because it was trained on Civil Comments.

Sentence Sets. Exp 1 uses 48 sentences covering 16 topics, each written three ways (promoting, reporting or arguing against abuse); 14 topics use an identical insult in all three (e.g. "Those fans are disgusting idiots and should be banned."). Exp 2 follows the template design of Dixon et al. [12]: 10 harmless templates (e.g. "My best friend is __.") filled with 15 identity words and four neutral control words (*tall, young, local, quiet*), giving 190 sentences. Exp 3 reuses these, and Exps 5 and 6 use all 238 sentences from Exps 1–2. Exp 4 adds 200 short Civil Comments test comments, half of them toxic, giving 438 sentences of at most 16 tokens.

Explainers. SHAP [6] and LIME [7] both treat each word as a feature, using a tokenizer that keeps punctuation attached to its word so that both score the same units. Exp 6 and LIME in Exps 1–2 explain only `original`; all other Detoxify runs explain both models. SHAP uses the Partition explainer on toxic log-odds, with `max_evals` = 500 (300 in Exp 2). Every SHAP run on Detoxify passed a check that its values sum to the model's output within 0.01. LIME explains the toxic probability instead, so its values are not compared with SHAP's in size. LIME uses 5,000 samples in Exps 3–5 and 2,000 in Exps 1–2. SHAP replaces removed words with "...", except in Exp 4, which deletes them.

References. We use two references. The *counterfactual effect* (Exp 3), following Garg et al. [21], is an identity sentence's log-odds minus the mean log-odds of its template's four control sentences; because only the identity word differs, the whole change is attributable to it. The *exact Shapley values* (Exp 4) are computed on the transparent filter from every subset of a sentence's words. SHAP (Partition) and LIME are run on this filter as a black box.

Metrics. Exp 2 reports the identity word's share of all absolute SHAP attribution. Faithfulness to the counterfactual effect (Exp 3) is measured per sentence and per term with Spearman's ρ, sign agreement and mean absolute error (MAE) in log-odds. For LIME, we use only ρ and sign agreement. Against exact Shapley values (Exp 4), we report the median ρ over sentences, top-word match and MAE, separately for sentences with low and high word-pair interaction (split at the median). The deletion test (Exp 5), adapted from ERASER [16], takes the top $k = 1, 2, 3$ words from SHAP, LIME or at random (averaged over 10 draws). *Comprehensiveness* is the drop in log-odds when these words are deleted (higher is better), and *sufficiency* is the drop when only they are kept (lower is better), both averaged over k (AOPC). For stability (Exp 6), LIME is run 10 times per sentence at 500 and 5,000 samples. We record how often the top word changes, the mean Spearman ρ between runs and LIME's R^2. *Signal strength* is the distance between a sentence's log-odds and SHAP's base value. We use AUC to test whether low R^2, SHAP–LIME disagreement or a weak signal can flag the 25% of identity sentences with the largest Exp 3 error.

IV. Results

Faithfulness to the References. In the identity-term swap (Exp 2), identity sentences scored far higher than control sentences (mean toxicity 0.124 vs. 0.001 for `original`), SHAP placed 46% (`original`) and 60% (`unbiased`) of its absolute attribution on the identity word, and SHAP and LIME chose the same top word in 78 of 88 sentences. Table I summarises every faithfulness result. In the counterfactual test (Exp 3), the true effect of identity term t and SHAP's estimate of it are

$$\Delta_t = \ell(x_t) - \overline{\ell(x_c)}, \qquad \hat{\Delta}_t = \phi_t - \overline{\phi_c}, \tag{1}$$

where ℓ is the filter's log-odds, ϕ is a SHAP value and the bars denote the mean over the four control-word versions x_c of the same template. Fig. 1 plots $\hat{\Delta}_t$ against Δ_t. SHAP ranks the effects almost perfectly ($\rho = 0.96$ for `original`, 0.92 for `unbiased`; 0.98 and 0.95 per term) and gives the correct sign in 98% and 93% of sentences. Its magnitudes are less accurate: the mean absolute error is 0.94 and 0.72 log-odds, and 83% and 75% of points lie above $y = x$, so SHAP tends to overstate the effect.

Under `unbiased`, the identity word's counterfactual effect rises only from 3.36 to 3.60 log-odds, while SHAP's estimate of it falls from 4.16 to 3.84.

Deleting removed words or using the mask token instead of "..." barely changes the ranking ($\rho \geq 0.97$ between baselines) but lowers SHAP's error from 0.94 log-odds ("...") to 0.84 (deletion) and 0.70 (mask token) for `original` (0.72, 0.58 and 0.44 for `unbiased`).

On the transparent filter (Exp 4), SHAP's Partition approximation matched the exact Shapley values almost perfectly (median $\rho = 0.993$, MAE 0.022 log-odds), and LIME's top word matched the exact one in 96% of runs. The filter's own coefficients are positive for 14 of the 15 identity words (e.g. gay $+2.83$, muslim $+1.90$ log-odds) and negative for all four control words (-0.26 to -0.39).

Deletion Test. Deleting the top-ranked words lowered the toxic log-odds far more than deleting random words (Table I; 4.02, 4.06 and 0.95 for `unbiased`). The gap was largest on the Exp 1 counter and report sentences and close to zero on control sentences. Sufficiency was negative for both explainers but positive for random words, so the top-ranked words alone scored higher than the full sentence, most strongly in report sentences (-3.4).

Where the Explainers Break. In the stability test (Exp 6), LIME's top word changes between seeds in 12% of sentences at 500 samples and in 4% at 5,000. Each sentence's instability (1 minus the mean Spearman ρ between runs) is unrelated to its signal strength $s = |\ell(x) - \phi_0|$, where ϕ_0 is SHAP's base value ($|\rho| < 0.05$). However, signal strength does predict

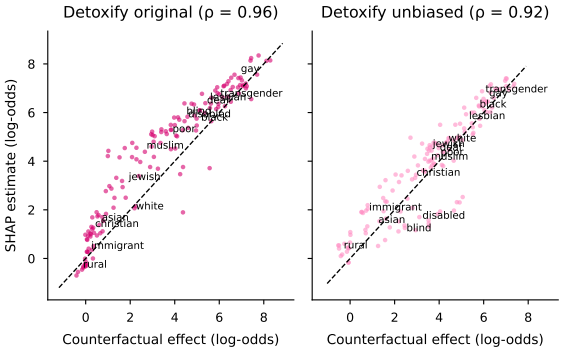


Fig. 1. SHAP's estimate $\hat{\Delta}_t$ vs. the counterfactual effect Δ_t for each identity sentence ($n = 150$ per model), in log-odds. Labels mark each term's mean; dashed line: $y = x$.

TABLE I
FAITHFULNESS OF SHAP AND LIME TO EACH REFERENCE

Measure	Exp	SHAP	LIME	Random
ρ with counterfactual, `original`	3	0.96	0.97	–
ρ with counterfactual, `unbiased`	3	0.92	0.97	–
ρ with exact Shapley, low interaction	4	1.00	1.00	–
ρ with exact Shapley, high interaction	4	0.99	0.98	–
Comprehensiveness ↑, `original`	5	3.09	3.05	0.35
Sufficiency ↓, `original`	5	−2.43	−2.65	1.30
Top word changes across seeds	6	–	4%	–

how well LIME's surrogate fits ($\rho = 0.73$ with R^2): at 5,000 samples the mean R^2 is 0.41 for the weaker half of sentences against 0.77 for the stronger half, and LIME's top word matches SHAP's in 86% against 95%.

Among cues that need no reference, weak signal strength best identified the 25% of identity sentences with the largest SHAP error $|\Delta_t - \hat{\Delta}_t|$ (AUC 0.80; 0.5 is chance), followed by low LIME R^2 (0.64), while SHAP–LIME disagreement did no better than chance (0.46).

V. DISCUSSION

Ranks Are Trustworthy, Magnitudes Are Not. SHAP ordered the identity effects almost perfectly ($\rho = 0.96$ and 0.92), and the ordering holds across terms as well as across sentences (0.98 and 0.95). Its magnitudes were less reliable: MAE was 0.94 and 0.72, mostly overstated, and this error shifted by up to 0.28 with the removal baseline while the ranking did not ($\rho \geq 0.97$). SHAP can tell an auditor *which* terms to examine, not *how large* the bias is. Attribution summaries can mislead outright: the identity word's share of $|\phi|$ rose from 46% to 60% under `unbiased`, suggesting the "debiased" model leans more on identity, yet its counterfactual effect rose by only 0.23 log-odds and SHAP's estimate of that effect fell.

Removal Is Not Neutral. Negative sufficiency means the top-ranked words alone scored as more toxic than the full sentence, most of all in report sentences (-3.4): stripped of its context, a sentence reporting abuse becomes a bare insult. Such fragments are out of distribution for the filter [17], so deletion metrics [16] partly measure the filter's reaction to fragments rather than explanation quality. Near-zero comprehensiveness on control sentences shows they separate signal from no signal.

Agreement Is Not Validation. SHAP and LIME chose the same top word in 78 of 88 sentences, but both explain by removing words and so share blind spots: their disagreement found SHAP's worst errors no better than chance (AUC 0.46). LIME's seed stability is similarly reassuring in the wrong way, since it stays stable where its surrogate fits poorly (mean R^2 0.41 on the weaker half of signals vs. 0.77). The cue that worked was signal strength (AUC 0.80). Attribution therefore suits *triage*, flagging strong identity effects for human review, but not *certifying* a filter as fair: its errors concentrate on weak signals, which is where a costly false "no bias found" would arise.

"Debiased" Is Not Identity-Insensitive. `unbiased` was trained to reduce bias aggregated over real comments [20], not to score a sentence identically when its identity word is swapped, so it can meet one fairness definition and fail ours. Transparency alone does not fix this either: our logistic filter, only modestly less accurate (AUC 0.892 vs. 0.928 and 0.971), learned the same bias from its data (gay $+2.83$) and simply makes it visible [8].

VI. CONCLUSION AND LIMITATIONS

We tested whether SHAP and LIME can support identity-term bias audits of a pretraining data filter. Both rank identity effects faithfully against counterfactual swaps ($\rho \geq 0.92$) and exact Shapley values ($\rho \geq 0.97$), and their top-ranked words matter far more than random ones (comprehensiveness 3.09 vs. 0.35). However, faithfulness is not uniform: SHAP overstates effect sizes, its magnitudes shift with the removal baseline, and LIME's surrogate fits poorly on weak signals. We therefore recommend that auditors use attribution as a triage tool, to flag suspicious filter behaviour for further review, rather than as a certification of fairness. In practice: (1) treat weak signal strength or low LIME R^2 as a warning that an explanation may be wrong; (2) do not treat SHAP–LIME agreement or seed stability as validation; (3) confirm any attribution finding with a counterfactual swap before drawing conclusions.

Limitations. Our scope is purposely narrow: we use short English template sentences with single-word identity substitutions, which do not capture dialect-level bias (e.g. AAE [3]) where no single word carries the signal. We test one filter family (Detoxify `original` and `unbiased`), and the two models differ in architecture (BERT vs. RoBERTa) as well as debiasing objective, so architectural effects cannot be fully separated. The transparent filter in Exp 4 is our own construction, not a real pipeline filter. Our ground-truth references, counterfactual effects and exact Shapley values, exist only because we controlled the inputs; a real auditor facing an opaque commercial filter would not have these. Future audits should extend to real web text (C4, Common Crawl), the quality filter of [2] and human rationale benchmarks such as HateXplain (surveyed in [19]).

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